



Applications: Sequences

You can use sequences to solve problems that involve number patterns.

A sequence may be arithmetic, geometric, or neither. For example, the sequence $1, -2, 3, -4, 5, \dots$ is neither arithmetic nor geometric, but you can examine the pattern and predict that the 6th term is -6 .

Determining Whether a Sequence Is Arithmetic, Geometric, or Neither

EXAMPLE 1

Decide whether the sequence is arithmetic, geometric, or neither.

A $-\frac{3}{2}, -\frac{3}{8}, -\frac{3}{32}, -\frac{3}{128}, \dots$

SOLUTION

Is it arithmetic?

Test for a common difference by subtracting.

$$b_2 - b_1 = -\frac{3}{8} - \left(-\frac{3}{2}\right) = \frac{9}{8}$$

$$b_3 - b_2 = -\frac{3}{32} - \left(-\frac{3}{8}\right) = \frac{9}{32}$$

There is no common difference, so it is not arithmetic.

Is it geometric?

Test for a common ratio by dividing.

$$b_2 \div b_1 = -\frac{3}{8} \div \left(-\frac{3}{2}\right) = -\frac{3}{8} \cdot \left(-\frac{2}{3}\right) = \frac{1}{4}$$

$$b_3 \div b_2 = -\frac{3}{32} \div \left(-\frac{3}{8}\right) = -\frac{3}{32} \cdot \left(-\frac{8}{3}\right) = \frac{1}{4}$$

$$b_4 \div b_3 = -\frac{3}{128} \div \left(-\frac{3}{32}\right) = -\frac{3}{128} \cdot \left(-\frac{32}{3}\right) = \frac{1}{4}$$

The common ratio is $\frac{1}{4}$. The sequence is geometric.

► **Remember** In an arithmetic sequence, the difference between consecutive terms is the common difference.

In a geometric sequence, the ratio between consecutive terms is the common ratio.

B 13, -8, -29, -50, ...

SOLUTION

Is it arithmetic?

Test for a common difference by subtracting.

$$z_2 - z_1 = -8 - 13 = -21$$

$$z_3 - z_2 = -29 - (-8) = -21$$

$$z_4 - z_3 = -50 - (-29) = -21$$

The sequence is arithmetic.

C 6, 7, 9, 12, ...

SOLUTION

Is it arithmetic?

Test for a common difference by subtracting.

$$t_2 - t_1 = 7 - 6 = 1$$

$$t_3 - t_2 = 9 - 7 = 2$$

There is no common difference, so it is not arithmetic.

Is it geometric?

Test for a common ratio by dividing.

$$t_2 \div t_1 = 7 \div 6 \approx 1.167$$

$$t_3 \div t_2 = 9 \div 7 \approx 1.286$$

There is no common ratio, so it is not geometric.

The sequence is neither arithmetic nor geometric. ■

EXAMPLE 2

Decide whether the sequence is arithmetic, geometric, or neither. Find the common difference or the common ratio, if there is one. Use the sequence to answer the question.

- A** A display of soup cans in the grocery store is stacked so that the top layer has 1 can, the next layer is 2 by 2 and has 4 cans, the next layer is 3 by 3 and has 9 cans, and the next layer is 4 by 4 with 16 cans. How many cans are in the layer below that?

SOLUTION

Write the sequence of numbers.

$$1, 4, 9, 16, \dots$$

Is it arithmetic?

Test for a common difference by subtracting.

$$4 - 1 = 3 \qquad 9 - 4 = 5$$

There is no common difference.

Is it geometric?

Test for a common ratio by dividing.

$$\frac{4}{1} = 4 \qquad \frac{9}{4} = 2.25$$

There is no common ratio.

The sequence is neither arithmetic nor geometric.

Use the pattern to find the next number in the sequence.

$$1^2 = 1 \qquad 2^2 = 4 \qquad 3^2 = 9 \qquad 4^2 = 16$$

The next layer will have 5^2 or 25 cans.

► **Think About It** To show that the test for a common difference or a common ratio fails, you only need to use two calculations. If the results are not the same, the test fails. However, to show that a common difference or a common ratio exists, you must test all the given numbers in the sequence.

- B** Suppose you find 1 penny on Monday, 2 pennies on Tuesday, 4 pennies on Wednesday, 8 pennies on Thursday, and 16 pennies on Friday. If this pattern continues, how many pennies will you find on Saturday?

SOLUTION

Write the sequence of numbers.

$$1, 2, 4, 8, 16, \dots$$

► **Think About It** What do the three dots following a list of numbers mean?

$$1, 2, 4, 8, 16, \dots$$

The dots show that the sequence continues in the same pattern without end.

Is it arithmetic?

Test for a common difference.

$$2 - 1 = 1 \qquad 4 - 2 = 2$$

There is no common difference.

Is it geometric?

Test for a common ratio.

$$\frac{2}{1} = 2 \qquad \frac{4}{2} = 2 \qquad \frac{8}{4} = 2 \qquad \frac{16}{8} = 2$$

The common ratio is 2, so the sequence is geometric.

Use the common ratio to find the next number in the sequence.

$$16 \cdot 2 = 32$$

The next number in the pattern is 32. You will find 32 pennies on Saturday.

- C** Makena adds \$15 every month to her savings jar. Every month, she counts the amount of money in the jar. What will her savings be after 6 months?

SOLUTION

Write the sequence of numbers.

$$15, 30, 45, 60, 75, \dots$$

Is it arithmetic?

Test for a common difference.

$$30 - 15 = 15 \quad 45 - 30 = 15 \quad 60 - 45 = 15 \quad 75 - 60 = 15$$

The common difference is \$15, so it is an arithmetic sequence.

Use the common difference to find the next number in the sequence.

$$75 + 15 = 90$$

After 6 months, Makena will have \$90. ■

Application: Money

EXAMPLE 3

Suppose someone offers to pay you 1 penny on the first day, 2 pennies the second day, and so on, doubling the number of pennies each day. (This pattern is the same pattern as in Example 2B.) What is the first day on which your pay would be at least \$10?

SOLUTION

Begin by changing \$10 to pennies. So the question is, on which day would you be paid at least 1000 pennies?

Step 1 Write what you know, and what you are looking for, in terms of the sequence rules.

You know the common ratio: $r = 2$. You know the first term: $a_1 = 1$.

You will be paid 1000 pennies on Day n , so you know the value of the n th term: $a_n = 1000$.

What you don't know is the value of n .

Step 2 Substitute the known values into the iterative rule.

$$a_n = a_1 \cdot r^{n-1} \quad \text{Iterative rule for a geometric sequence}$$

$$1000 = 1 \cdot 2^{n-1} \quad \text{Substitute.}$$

$$1000 = 2^{n-1} \quad \text{Simplify.}$$

Step 3 Use the guess-and-check strategy to solve for n .

$$\text{Try } n = 9: 2^{9-1} = 2^8 = 256 \quad \text{too low}$$

$$\text{Try } n = 10: 2^{10-1} = 2^9 = 512 \quad \text{too low}$$

$$\text{Try } n = 11: 2^{11-1} = 2^{10} = 1024 \quad \text{too high}$$

n is between 10 and 11. To determine if you should round up or down, think about what you are looking for: The first day, you earn at least 1000 pennies. On Day 10, your pay will be 512 pennies. On Day 11, your pay will be 1024 pennies, so the answer is Day 11. ■

► **Remember** When you know one term, a_n , of an arithmetic or a geometric sequence, use a recursive rule to find the next term.

For an arithmetic sequence:

$$a_n = a_{n-1} + d$$

For a geometric sequence:

$$a_n = r \cdot a_{n-1}$$

In both of these rules, a_n is any term after the first, called the n th term.

The previous term is a_{n-1} . The common difference is d , and the common ratio is r .

Application: Population Growth

EXAMPLE 4

A small town had a population of 968 in 2000. By the next year, the population had grown to 1007. In 2002, there were 1047 people, and the next year there were 1089.

A What kind of sequence is this?

SOLUTION

Write the sequence of numbers.

$$968, 1007, 1047, 1089, \dots$$

Step 1 Test for a common difference by subtracting.

$$1007 - 968 = 39 \qquad 1047 - 1007 = 40 \qquad 1089 - 1047 = 42$$

There is no common difference.

Step 2 Test for a common ratio by dividing.

$$\frac{1007}{968} \approx 1.04 \qquad \frac{1047}{1007} \approx 1.04 \qquad \frac{1089}{1047} \approx 1.04$$

There is a common ratio. This sequence is geometric.

Each year the population increases by about 4%.

► **Think About It** The ratios in Example 4A aren't exactly equal to each other, but they are pretty close, so a geometric sequence is a reasonable model.

B If this pattern continues, what will the population be after 15 years?

SOLUTION

Use the iterative rule, $a_n = a_1 \cdot r^{n-1}$, with the values you know.

$$a_n = a_1 \cdot r^{n-1}$$

$$a_{15} \approx 968 \cdot 1.04^{15-1} \quad \text{Substitute.}$$

$$a_{15} \approx 968 \cdot 1.04^{14} \quad \text{Simplify.}$$

$$a_{15} \approx 1676 \quad \text{Use a calculator to approximate the value.}$$

After 15 years, the population will be about 1676. ■

Application: Finance

EXAMPLE 5

The balance on a car loan is \$7600. The car owner makes a payment of \$200 every month.

A Write a sequence showing the balance for the first 5 months. What kind of sequence is this?

SOLUTION

The sequence is \$7600, \$7400, \$7200, \$7000, There is a common difference of $-\$200$, so this is an arithmetic sequence.

B When will the balance be less than \$600?

SOLUTION

Find the number of the term that is equal to \$600.

$$c_n = c_1 + (n - 1) \cdot d$$

$$600 = 7600 + (n - 1) \cdot -200 \quad \text{Substitute.}$$

$$600 = 7600 - 200n + 200 \quad \text{Distribute } -200.$$

$$-7200 = -200n \quad \text{Simplify.}$$

$$36 = n \quad \text{Divide both sides by } -200.$$

Notice that the first term, c_1 , is \$7600, and the balance after 1 payment is c_2 , or \$7400. So c_{36} is the balance after 35 payments, and $c_{36} = 600$. The balance will be less than \$600 after 36 payments. ■

Application: Physics

EXAMPLE 6

A ball is dropped from a height of 100 in. The ball bounces, each time reaching a lower and lower height. The height of each bounce is 80% of the height of the previous bounce.

- A** What height, rounded to the nearest inch, will the ball reach after bouncing 4 times?

SOLUTION

In this situation, the first height in this geometric sequence is 100 in., so $f_1 = 100$. The second height will be 80% of 100 in., and so forth. The height, after bouncing 4 times, is the fifth height in the sequence, f_5 .

Step 1 Write what you know, and what you are looking for, in sequence notation.

The common ratio is 80%, or 0.8, so $r = 0.8$, and you know $f_1 = 100$.

You are looking for the value of the fifth term, f_5 .

Step 2 Find f_n when $n = 5$.

$$f_n = f_1 \cdot r^{n-1}$$

$$f_5 = 100 \cdot 0.8^{5-1} \quad \text{Substitute.}$$

$$f_5 = 100 \cdot 0.8^4 \quad \text{Simplify.}$$

$$f_5 = 40.96 \quad \text{Use a calculator to evaluate.}$$

After bouncing 4 times, the ball will reach a height of almost 41 in.

► **Think About It** Because the height of each successive bounce is 80% of a positive number, the ball's bounce, in theory, will never reach a zero height. In practice, however, the ball loses energy and does eventually stop bouncing.

► **Think About It** To be sure of the pattern in a sequence, you may need to know more than just three terms.

- B** How many bounces will it take for the ball to reach a height of less than 1 in.?

SOLUTION

You need to solve for n when the height f_n is 1 in.

$$f_n = f_1 \cdot r^{n-1}$$

$$1 = 100 \cdot 0.8^{n-1} \quad \text{Substitute.}$$

Use the guess-and-check strategy to solve for n .

$$\text{Try } n = 23: 100 \cdot 0.8^{23-1} = 100 \cdot 0.8^{22} \approx 0.7379 \quad \text{too low}$$

$$\text{Try } n = 22: 100 \cdot 0.8^{22-1} = 100 \cdot 0.8^{21} \approx 0.9223 \quad \text{too low}$$

$$\text{Try } n = 21: 100 \cdot 0.8^{21-1} = 100 \cdot 0.8^{20} \approx 1.153 \quad \text{too high}$$

n is between 21 and 22, but only whole number values make sense in this situation. Think about what you are looking for: the number of bounces it will take for the ball to reach a height of less than 1 in. Because $f_{21} > 1$, you want the bounce number that corresponds with f_{22} . The bounce number is 1 less than n , so the answer is 21 bounces. ■

Actually, if you try this with a real ball, the ball would probably not bounce as many as 21 times. It would lose energy and stop bouncing before that.